



I trained a GPT on nothing but Shakespeare.

No pretraining. No borrowed tokenizer. No outside text.
Eleven million parameters, on one laptop.

THE RULE

Pretend the only text that ever existed was Shakespeare.

- **44 works, ~5 MB, 97 characters.** 38 plays, 154 sonnets, 5 poems — and not one other word.

- **Everything built from nothing.** The tokenizer, the model, the training loop — no imports from a pretrained world.

- **It reads one character at a time.** A comparison with fixed rules chose characters over word-fragments.

It was measuring it honestly.

- **Three works locked away on day one.** A history, a romance, a poem — never trained on, never looked at.

- **Every rule written down before the number.** Which run to release, what counts as memorising — fixed in advance, in the open.

- **A one-way door.** The held-out works could be scored exactly once. A score is only held out until you use it.

THE ONE NUMBER, ON WORKS IT NEVER READ

1.80 bits per character
on unseen Shakespeare

a coin-flip per character	1.00
GP-Thee-11M	1.80
best n-gram model (no network)	2.32
a good file compressor	2.60
the raw 97-character alphabet	6.60

It beats every predictor built without a neural network — yet reproduces almost none of the verse it trained on. The longest line of Shakespeare it can be made to recite is **25 characters**. It learned the *shape* of an edition, not the poetry.

WHAT IT IS, AND WHY

An in-character autocomplete for a world of only Shakespeare.

It can't answer a question or follow an instruction. Give it a speaker and a line and it plays the next part. I built it for one reason: to see whether a language model could be made, and measured, **honestly — every number checked before it was written.** It could.

Full build, code & logs → **github.com/shivamtiwari93/gp-thee**

The weights → **[Hugging Face: GP-Thee-11M](#)**

Eight-part write-up → **[in the repo](#)**